

MATH 180 Strategies of Problem Solving

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Outline

1

Board Shapes

- Young Diagrams
- Boundary Paths

2

Counting Tools

- Factorials
- Binomial Coefficients

3

Young Diagrams and Applications

- General Formula
- Applications
- Wrapping Up

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Young Diagrams and Applications

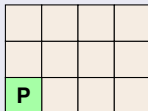
- General Formula
- Applications
- Wrapping Up

Question from last time

Question from last time

Puzzle

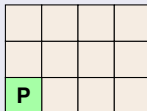
Start with a 3×4 Chomp board. How many different board shapes can occur during play?



Question from last time

Puzzle

Start with a 3×4 Chomp board. How many different board shapes can occur during play?



A detail to decide

Should we include the empty board after someone eats the poison?

Counting board shapes

Activity (10 mins)

Work with your neighbors.

- Find a systematic way to describe every possible shape.
- Count without repeating a shape.
- Explain how you know your list is complete.

Counting board shapes

Activity (10 mins)

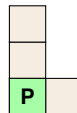
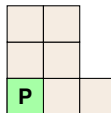
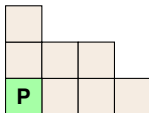
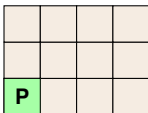
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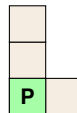
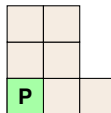
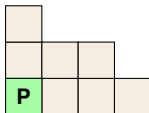
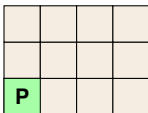
A useful goal

Replace the pictures with objects that are easier to count.

Some possible boards



Some possible boards



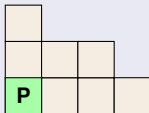
Question

How could we describe each board without drawing it?

Row lengths

Encoding

Read the row lengths from bottom to top.

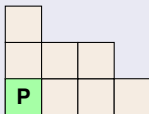


corresponds to $(4, 3, 1)$.

Row lengths

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Question

Which triples (a, b, c) describe possible boards inside a 3×4 rectangle?

The condition on the rows

Answer

A board inside a 3×4 rectangle has row lengths

$$4 \geq a \geq b \geq c \geq 0.$$

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Reason

A square cannot remain unless every square below it and every square to its left also remains.

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Terminology

Mathematicians call a shape of left-aligned rows with decreasing lengths a **Young diagram**.

Young diagrams in our orientation

Usual convention

Young diagrams are often drawn with the longest row on top.

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Our convention

We draw the longest row on the bottom so the poisoned square stays in the bottom-left, just as it did in Chomp.

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Our convention

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Same mathematical object

Turning the paper upside down changes the picture, but not the row lengths or the counting problem.

A first census

Warm-up

How many Young diagrams fit inside each rectangle?

Rectangle	$1 \times n$	2×2	2×3
Number of shapes	?	?	?

A first census

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Answers

$n + 1$, 6, 10.

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Question

What pattern might continue?

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- **Boundary Paths**

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A path around every shape

Boundary idea

Trace the boundary separating the remaining chocolate from the missing chocolate.

From paths to words

Record each step

Read from top-left to bottom-right. Write H for a step to the right and V for a step downward.

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Example

$HHVHVHV$

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Example

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What every word contains

Exactly four H 's and three V 's, in some order.

Why this correspondence works

Young diagram to word

Each Young diagram has one boundary path, so it produces one word.

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Word to Young diagram

A word with four H 's and three V 's traces one path across the rectangle. The squares on the bottom-left side form one Young diagram.

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Word to Young diagram

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Bijection

Every diagram gets one word, and every word gets one diagram. Therefore the two collections have the same size.

The counting problem has changed

Original question

How many Young diagrams fit in a 3×4 rectangle?

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Equivalent question

How many different arrangements of

H, H, H, H, V, V, V

are possible?

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Strategy

We first learn to count arrangements of objects that are all different.

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The multiplication principle

Two consecutive choices

If a first choice has a possibilities and each can be followed by b possibilities, then the pair of choices has ab possibilities.

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Example

Three shirts and four pairs of pants make

$$3 \cdot 4 = 12$$

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More than two choices

Multiply the number of possibilities at each stage.

Arranging different objects

Example

Four students will give presentations. How many speaking orders are possible?

Arranging different objects

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Fill the four positions

$$\underbrace{4}_{\text{first}} \cdot \underbrace{3}_{\text{second}} \cdot \underbrace{2}_{\text{third}} \cdot \underbrace{1}_{\text{fourth}} = 24.$$

Arranging different objects

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Four students will give presentations. How many speaking orders are possible?

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Why the numbers decrease

Once a student has a position, that student cannot be chosen again.

Factorials

Definition

For a positive integer n ,

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1.$$

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Examples

$$3! = 6, \quad 5! = 120, \quad 7! = 5040.$$

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Examples

$$3! = 6, \quad 5! = 120, \quad 7! = 5040.$$

Useful convention

$0! = 1$: there is exactly one way to arrange no objects, the empty order.

A factorial checkpoint

Think, then compare

Seven different books are placed in a row.

- 1 How many orders are possible?
- 2 How many orders put a particular book first?
- 3 How many orders put two particular books next to each other?

A factorial checkpoint

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Answers

$7!$, $6!$, $2 \cdot 6!$.

Coffee break (5 mins)

Why is $7!$ too large for arranging

H, H, H, H, V, V, V ?

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Repeated objects change the count

Pretend the letters have labels

Arrange

$$H_1, H_2, H_3, H_4, V_1, V_2, V_3.$$

Since these seven objects are different, there are $7!$ orders.

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Erase the labels

Every visible H/V word came from

$$4! \cdot 3!$$

labeled orders: the H labels and V labels can be rearranged invisibly.

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labeled orders: the H labels and V labels can be rearranged invisibly.
Therefore the corrected count is

$$\frac{7!}{4!3!} = 35.$$

A second route: choose positions

Seven empty positions

— — — — — — —

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Choose three positions for V

Once those positions are chosen, all four remaining positions must contain H .

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Same question

How many ways can we choose three positions from seven positions?

Binomial coefficients

Notation

Read $\binom{n}{k}$ as “ n choose k .”

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Formula

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Why the binomial formula works

First count ordered selections

Choose k objects in order:

$$n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

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Then forget the order

Each group of k objects appears in $k!$ different orders.

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First count ordered selections

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Then forget the order

Each group of k objects appears in $k!$ different orders.

Divide out the overcount

$$\binom{n}{k} = \frac{n!}{(n-k)!} \cdot \frac{1}{k!} = \frac{n!}{k!(n-k)!}.$$

Example: choosing a committee

Problem

A class has eight students. How many three-student committees are possible?

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Calculation

$$\binom{8}{3} = \frac{8!}{3!5!} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56.$$

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Important distinction

A committee has no first, second, or third position. If the jobs were different, we would count an ordered selection instead.

Example: binary messages

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How many eight-bit messages contain exactly three 1's?

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Choose positions

Choose which three of the eight positions contain a 1. The other five positions contain 0.

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Choose positions

Choose which three of the eight positions contain a 1. The other five positions contain 0.

Answer and connection

$$\binom{8}{3} = 56.$$

This is the same repeated-letter structure as arranging H 's and V 's.

Choosing a group or its complement

Choose who serves

Choosing k people from n determines the committee.

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Choose who does not serve

The same decision also chooses the other $n - k$ people.

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Symmetry

$$\binom{n}{k} = \binom{n}{n-k}.$$

For boundary words, choose the V positions or choose the H positions.

A useful recurrence

Question

How many k -person committees can be chosen from n people if we focus on one particular person, Alex?

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Two cases

- Alex serves: choose the other $k - 1$ people from $n - 1$.

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Two cases

- Alex serves: choose the other $k - 1$ people from $n - 1$.
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Pascal's identity

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

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Back to the 3×4 board

Boundary words

Each Young diagram corresponds to one word with four H 's and three V 's.

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Interpretation

There are 35 diagrams including the empty diagram. There are 34 playable Chomp positions in which the poison remains.

The general formula

An $m \times n$ rectangle

From top-left to bottom-right, every boundary word has m downward steps and n rightward steps.

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Choose either type of step

$$\text{Number of Young diagrams} = \binom{m+n}{m} = \binom{m+n}{n}.$$

The general formula

An $m \times n$ rectangle

From top-left to bottom-right, every boundary word has m downward steps and n rightward steps.

Choose either type of step

$$\text{Number of Young diagrams} = \binom{m+n}{m} = \binom{m+n}{n}.$$

What makes the formula possible

A diagram is completely determined by an arrangement of two kinds of steps.

Checking the formula

Rectangle	Formula	Number of diagrams
$1 \times n$	$\binom{n+1}{1}$	$n + 1$
2×2	$\binom{4}{2}$	6
2×3	$\binom{5}{2}$	10
3×4	$\binom{7}{3}$	35
4×5	$\binom{9}{4}$	126

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Question

Why do $m \times n$ and $n \times m$ rectangles give the same answer?

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Application 1: shortest routes

Problem

A delivery robot must travel six blocks east and four blocks north. It never moves west or south. How many shortest routes are possible?

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Answer and Young diagram connection

$$\binom{10}{4} = 210.$$

Reflecting the route vertically makes it the boundary of a Young diagram inside a 4×6 rectangle.

Application 2: merging two lanes

Problem

Five cars wait in lane A and seven cars wait in lane B . Cars cannot pass within a lane. How many merge orders are possible?

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Encoding

Write A whenever the next car comes from lane A , and B otherwise. The order of the cars within each lane is already fixed.

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Five cars wait in lane A and seven cars wait in lane B . Cars cannot pass within a lane. How many merge orders are possible?

Encoding

Write A whenever the next car comes from lane A , and B otherwise. The order of the cars within each lane is already fixed.

Answer

$$\binom{12}{5} = 792.$$

Application 3: final grades

Model

Gandalf, Dumbledore, Obi-Wan Kenobi, Merlin, and Tim the Enchanter all took Math 180 last year. You heard from a friend that in terms of final letter grades

$$\text{Tim} \leq \text{Kenobi} \leq \text{Dumbledore} \leq \text{Merlin} \leq \text{Gandalf}.$$

How many possible letter grade combinations could have occurred?

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Shape

Correct answers form a Young diagram in a 5×5 square.

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Shape

Correct answers form a Young diagram in a 5×5 square.

Answer

$$\binom{10}{5} = 252.$$

One formula, several stories

Problem	First symbol	Second symbol
Young diagram	rightward step	downward step
Shortest route	east move	north move
Merge	lane A car	lane B car
Work schedule	program A job	program B job

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Shared structure

In every case, an outcome is determined by arranging m symbols of one kind and n symbols of another kind.

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Question of the Day

Puzzle

A project has five design tasks that must remain in their given order and eight testing tasks that must remain in their given order. The team may interleave design and testing. How many schedules are possible?

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Explain, do not only calculate

Give the binomial coefficient, its value, and a correspondence between schedules and Young diagrams.

Final thoughts

Reflection

In your own words, explain one of these ideas:

Final thoughts

Reflection

In your own words, explain one of these ideas:

- why factorials count arrangements of different objects;

Final thoughts

Reflection

In your own words, explain one of these ideas:

- why factorials count arrangements of different objects;
- why repeated objects lead to binomial coefficients;

Final thoughts

Reflection

In your own words, explain one of these ideas:

- why factorials count arrangements of different objects;
- why repeated objects lead to binomial coefficients;
- how a boundary word counts Young diagrams and real-world outcomes.

Submit your response by **Tonight!**